Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For

Microsoft Azure AI Fundamentals (AI-900)

TGS Ref No: TGS-2023021100

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
9.0	Legacy	Legacy trainer deck and lab coverage	Tertiary Infotech Academy Pte Ltd
10.0	2 September 2026	Visual rebuild; Microsoft Foundry terminology; updated 10-lab package; aligned WA/PP	Dr Alfred Ang

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Course status note

Microsoft retired the AI-900 certification exam on 30 June 2026. This WSQ course retains the registered title and outcomes and teaches the Azure AI fundamentals and Microsoft Foundry capabilities represented by the final published AI-900 blueprint. No mock-test material is included.

Learning outcomes

- LO1: Implement AI applications with Microsoft Azure services.
- LO2: Deploy AI workflows on Azure.
- LO3: Identify and report issues with AI applications and anomaly-detection evidence.
- LO4: Maintain data interoperability and perform data cleaning with Azure knowledge mining.

TSC knowledge and abilities

- K1: Background and basics of AI
- K2: Fundamental concepts and methods of statistics and programming for data science
- K3: Mathematics and computing theories
- K4: Machine learning techniques and applications
- K5: Principles of data management
- K6: Software development methodologies
- K7: Types of AI applications
- K8: AI implementation procedures
- K9: Interoperability of data
- K10: Data cleaning techniques
- K11: Principles of clean data sets
- A1: Implement AI applications in collaboration with technology service providers
- A2: Deploy AI workflows according to plan
- A3: Identify and report issues with AI applications and data collected
- A4: Maintain data interoperability during the development of AI
- A5: Perform data cleaning techniques

Course access and lab conventions

1. Sign in to https://lms-tms.tertiaryinfotech.com/ with the account provided by the academy.
2. Open Microsoft Azure AI Fundamentals and verify the course code is TGS-2023021100.
3. Download the current learner slides, Learner Guide and Labs folder.
4. Work inside one lab folder at a time and save evidence only in its evidence subfolder.
5. Never place keys, tokens, personal data or confidential documents in screenshots or files.
6. Delete or stop billable Azure resources after the trainer verifies the evidence.

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Source and terminology notes

The v9 deck is preserved as the coverage floor. Product names and UI guidance were refreshed against the pinned MicrosoftLearning repositories in SOURCES.md. Full legacy explanations are summarized below; obsolete provisioning paths are replaced by current mechanisms and migration notes.

Topic 1 — Get started with artificial intelligence

Describe Artificial Intelligence workloads and considerations...

Legacy source slide 20; aligned new slide is recorded in slide_map.json.

- AI-900 Azure AI Fundamentals
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

What is AI?

Legacy source slide 21; aligned new slide is recorded in slide_map.json.

- Simply put, AI is the creation of software that imitates human behaviors and capabilities. Natural language processing - The capability for a computer to interpret written or…
- Key workloads include: Machine learning - This is often the foundation for an AI system, and is the way we "teach" a computer model to make predictions and draw conclusions……
- Anomaly detection - The capability to automatically detect errors or unusual activity in a system.
- Computer vision - The capability of software to interpret the world visually through cameras, video, and images.

Microsoft Azure provides the Azure Machine Learning service - a...

Legacy source slide 22; aligned new slide is recorded in slide_map.json.

- Machine Learning in Microsoft Azure
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

A machine learning based technique that analyzes data over time and...

Legacy source slide 23; aligned new slide is recorded in slide_map.json.

- Anomaly Detection
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Computer Vision is an area of AI that deals with visual processing....

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- Computer Vision
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Image classification involves training a machine learning model to...

Legacy source slide 25; aligned new slide is recorded in slide_map.json.

- Image Classification
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Object detection machine learning models are trained to classify...

Legacy source slide 26; aligned new slide is recorded in slide_map.json.

- Object Detection
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Semantic segmentation is an advanced machine learning technique in...

Legacy source slide 27; aligned new slide is recorded in slide_map.json.

- Semantic Segmentation
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

You can create solutions that combine machine learning models with...

Legacy source slide 28; aligned new slide is recorded in slide_map.json.

- Image Analysis
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Face detection is a specialized form of object detection that...

Legacy source slide 29; aligned new slide is recorded in slide_map.json.

- Face Detection
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.

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- Control: define the threshold or failure that routes to human review.

Optical character recognition is a technique used to detect and read...

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- OCR
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Natural Language Processing

Legacy source slide 31; aligned new slide is recorded in slide_map.json.

- Natural language processing (NLP) is the area of AI that deals with creating software that understands written and spoken language NLP enables you to create software that can:…
- Interpret spoken language, and synthesize speech responses.
- Automatically translate spoken or written phrases between languages.
- Interpret commands and determine appropriate actions.

Natural Language Processing in Microsoft Azure

Legacy source slide 32; aligned new slide is recorded in slide_map.json.

- Normalized text produces typed outputs such as language, sentiment, entities or redacted spans.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Microsoft AI Principles

Legacy source slide 33; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Microsoft AI Principles

Legacy source slide 34; aligned new slide is recorded in slide_map.json.

- Fairness: AI systems shouldn't make decisions that discriminate against or support bias of a group or individual. Accountability: People must be accountable for how AI systems…

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- Reliability and Safety: AI systems should respond safely to new situations and potential manipulation. Transparency: AI systems should have explanations so users can understand…
- Privacy and Security: AI systems should be secure and respect data privacy.
- Inclusiveness: AI systems should empower everyone and engage people.

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Legacy source slide 35; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Legacy source slide 36; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Legacy source slide 37; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Legacy source slide 38; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Legacy source slide 39; aligned new slide is recorded in slide_map.json.

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- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Conversational AI

Legacy source slide 40; aligned new slide is recorded in slide_map.json.

- Conversational AI is the term used to describe solutions where AI agents participate in conversations with humans. Reservation systems for restaurants, airlines, cinemas, and…
- Most commonly, conversational AI solutions use bots to manage dialogs with users. Health care consultations and self-diagnosis.
- These dialogs can take place through web site interfaces, email, social media platforms, messaging systems, phone calls, and other channels. Home automation and personal…
- Bots can be the basis of AI solutions for: Customer support for products or services.

To create conversational AI solutions on Microsoft Azure, you can...

Legacy source slide 41; aligned new slide is recorded in slide_map.json.

- Conversational AI in Microsoft Azure
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Try out the Azure Conversational AI below...

Legacy source slide 42; aligned new slide is recorded in slide_map.json.

- Activity: Conversational AI
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

OpenAI makes its AI models available to developers to build powerful...

Legacy source slide 43; aligned new slide is recorded in slide_map.json.

- What is Generative AI
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure OpenAI

Legacy source slide 44; aligned new slide is recorded in slide_map.json.

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- Azure OpenAI is available for Azure users and consists of four components: Pre-trained generative AI models Customization capabilities; the ability to fine-tune AI models with…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure AI evidence 045

Legacy source slide 45; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure AI evidence 046

Legacy source slide 46; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Microsoft Foundry model playground

Legacy source slide 47; aligned new slide is recorded in slide_map.json.

- In the Microsoft Foundry model playground, you can build AI models and deploy them for public consumption in software applications.
- Azure OpenAI's capabilities are made possible by specific generative AI models.
- These Azure OpenAI models fall into a few main families: GPT-4 current chat model family Codex Embeddings image generation model
- Control: define the threshold or failure that routes to human review.

GitHub Copilot

Legacy source slide 48; aligned new slide is recorded in slide_map.json.

- OpenAI partnered with GitHub to create GitHub Copilot, which they call an AI pair programmer.
- GitHub Copilot integrates the power of OpenAI Codex into a plugin for developer environments like Visual Studio Code.
- Once the plugin is installed and enabled, you can start writing your code, and GitHub Copilot starts automatically suggesting the remainder of the function based on code…
- Control: define the threshold or failure that routes to human review.

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For example, we have only a function name in the file, and the gray...

Legacy source slide 49; aligned new slide is recorded in slide_map.json.

- GitHub Copilot
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

GitHub Copilot offers multiple suggestions for code completion,...

Legacy source slide 50; aligned new slide is recorded in slide_map.json.

- GitHub Copilot
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

OpenAI's NLP Capabilities

Legacy source slide 51; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

In addition to natural language capabilities, generative AI models...

Legacy source slide 52; aligned new slide is recorded in slide_map.json.

- image generation model
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamental AI Concepts

Legacy source slide 53; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/get-started-ai-fundamentals/8b-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.get-started-with-art…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of machine learning

Legacy source slide 54; aligned new slide is recorded in slide_map.json.

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- https://learn.microsoft.com/en-us/training/modules/fundamentals-machine-learning/11-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.get-started-with-a…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of Azure AI services

Legacy source slide 55; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/fundamentals-azure-ai-services/7-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.get-started-with-a…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Topic 2 — Visual tools for machine learning

What is Machine Learning

Legacy source slide 57; aligned new slide is recorded in slide_map.json.

- Machine learning is a technique that uses mathematics and statistics to create a model that can predict unknown values.
- Mathematically, you can think of machine learning as a way of defining a function (let's call it f) that operates on one or more features of something (which we'll call x) to…
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Types of Machine Learning

Legacy source slide 58; aligned new slide is recorded in slide_map.json.

- There are two general approaches to machine learning: supervised and unsupervised machine learning. The unsupervised machine learning approach starts with a dataset without…
- In both approaches, you train a model to make predictions. One type of unsupervised machine learning task is clustering.
- The supervised machine learning approach requires you to start with a dataset with known label values.
- Two types of supervised machine learning tasks include regression and classification.

Azure Machine Learning Studio

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- Azure Machine Learning Studio is the web portal for data scientist developers in Azure Machine Learning.
- The studio combines no-code and code-first experiences for an inclusive data science platform.
- The studio offers multiple authoring experiences depending on the type project and the level of user experience.
- Control: define the threshold or failure that routes to human review.

Azure Machine Learning Studio

Legacy source slide 60; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure Machine Learning Compute

Legacy source slide 61; aligned new slide is recorded in slide_map.json.

- In Azure Machine Learning studio, you can manage the compute targets for your data science activities. Attached Compute: Links to existing Azure compute resources, such as…
- There are four kinds of compute resource you can create: Compute Instances: Development workstations that data scientists can use to work with data and models.
- Compute Clusters: Scalable clusters of virtual machines for on-demand processing of experiment code.
- Inference Clusters: Deployment targets for predictive services that use your trained models.

You can write and run your own code in managed Jupyter Notebook…

Legacy source slide 62; aligned new slide is recorded in slide_map.json.

- Azure ML Studio Notebook
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

You can use the designer to train and deploy machine learning models…

Legacy source slide 63; aligned new slide is recorded in slide_map.json.

- Azure Machine Learning Designer
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Azure Machine Learning Designer

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- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Pipelines

Legacy source slide 65; aligned new slide is recorded in slide_map.json.

- Pipelines let you organize, manage, and reuse complex machine learning workflows across projects and users.
- A pipeline starts with the dataset from which you want to train the model.
- Each time you run a pipeline, the configuration of the pipeline and its results are stored in your workspace as a pipeline job.
- Control: define the threshold or failure that routes to human review.

Pipelines

Legacy source slide 66; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Components

Legacy source slide 67; aligned new slide is recorded in slide_map.json.

- An Azure Machine Learning component encapsulates one step in a machine learning pipeline.
- You can think of a component as a programming function and as a building block for Azure Machine Learning pipelines.
- In a pipeline project, you can access data assets and components from the left panel's Asset Library tab.
- Control: define the threshold or failure that routes to human review.

Components

Legacy source slide 68; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.

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- Control: define the threshold or failure that routes to human review.

You can create data assets on the Data page from local files, a...

Legacy source slide 69; aligned new slide is recorded in slide_map.json.

- Datasets
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

An Azure Machine Learning (ML) job executes a task against a...

Legacy source slide 70; aligned new slide is recorded in slide_map.json.

- Azure Machine Learning Jobs
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure Machine Learning Jobs

Legacy source slide 71; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure Machine Learning Jobs

Legacy source slide 72; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Regression

Legacy source slide 73; aligned new slide is recorded in slide_map.json.

- Regression is a form of machine learning that is used to predict a numeric label based on an item's features.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
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Regression Example

Legacy source slide 74; aligned new slide is recorded in slide_map.json.

- Prepare data: Identify the features and label in a dataset. Test the machine learning model for performance using the validation data set.
- Pre-process, or clean and transform, the data as needed. Evaluate performance: Compare how close the model's predictions are to the known labels.
- Train model: Split the data into two groups, a training and a validation set. Deploy a predictive service: After you train a machine learning model, you need to convert the…
- Train a machine learning model using the training data set. Then you can deploy the model as an application on a server or device so that others can use it.

Regression Example

Legacy source slide 75; aligned new slide is recorded in slide_map.json.

- A regression model maps features to a numeric label; MAE and RMSE quantify prediction error on held-out data.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Inference Pipeline

Legacy source slide 76; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Automatic Price Prediction Regression

Legacy source slide 77; aligned new slide is recorded in slide_map.json.

- A regression model maps features to a numeric label; MAE and RMSE quantify prediction error on held-out data.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Next to the pipeline name on the left, select the button >> to...

Legacy source slide 78; aligned new slide is recorded in slide_map.json.

- Add and Explore a Dataset
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.

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- Control: define the threshold or failure that routes to human review.

You typically apply data transformations to prepare the data for...

Legacy source slide 79; aligned new slide is recorded in slide_map.json.

- Add Data Transformations
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Select the Select Columns in Dataset module, and in its Settings...

Legacy source slide 80; aligned new slide is recorded in slide_map.json.

- Select the Select Column
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Clean Missing Data

Legacy source slide 81; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Normalize Data

Legacy source slide 82; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Run the Pipeline

Legacy source slide 83; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Add Training Modules

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- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Run the Training Pipeline

Legacy source slide 85; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Evaluate a Regression Model

Legacy source slide 86; aligned new slide is recorded in slide_map.json.

- A regression model maps features to a numeric label; MAE and RMSE quantify prediction error on held-out data.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Model Performance Metrics

Legacy source slide 87; aligned new slide is recorded in slide_map.json.

- Mean Absolute Error (MAE): The average difference between predicted values and true values.
- Root Mean Squared Error (RMSE): The square root of the mean squared difference between predicted and true values.
- Coefficient of Determination (R2): This metric is more commonly referred to as R-Squared, and summarizes how much of the variance between predicted and true values is explained…
- The closer to 1 this value is, the better the model is performing.

Create an Inference Pipeline

Legacy source slide 88; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.

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- Control: define the threshold or failure that routes to human review.

Deployment

Legacy source slide 89; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Testing

Legacy source slide 90; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Testing

Legacy source slide 91; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Activity: Regression

Legacy source slide 92; aligned new slide is recorded in slide_map.json.

- Import the Boston dataset from internet https://raw.githubusercontent.com/selva86/datasets/master/BostonHousing.csv Create a Predictive Regression Model for Housing Price based…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Classification is a form of machine learning that is used to predict...

Legacy source slide 93; aligned new slide is recorded in slide_map.json.

- Classification
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Classification Example

Legacy source slide 94; aligned new slide is recorded in slide_map.json.

- A classifier maps a feature vector to a categorical label and exposes confidence or class probability for validation.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Before you can train a model, you typically need to apply some...

Legacy source slide 95; aligned new slide is recorded in slide_map.json.

- Add Data Transformations
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Select the Normalize Data module and view its settings, noting that...

Legacy source slide 96; aligned new slide is recorded in slide_map.json.

- Normalize Data
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

To apply your data transformations, you need to run the pipeline as...

Legacy source slide 97; aligned new slide is recorded in slide_map.json.

- Run the Pipeline
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Add Training Modules

Legacy source slide 98; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Run the Training Pipeline

Legacy source slide 99; aligned new slide is recorded in slide_map.json.

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- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Create an Inference Pipeline

Legacy source slide 100; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

When the experiment run has finished, select the Evaluate Model...

Legacy source slide 101; aligned new slide is recorded in slide_map.json.

- Confusion Matrix
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Confusion Matrix

Legacy source slide 102; aligned new slide is recorded in slide_map.json.

- The confusion matrix is a tool used to assess the quality of a classification model's predictions. False Negative: The model predicts the patient does not have diabetes, but…
- It compares predicted labels against actual labels. True Negative: The model predicts the patient does not have diabetes, and the patient actually does not have diabetes.
- True Positive: The model predicts the patient has diabetes, and the patient does actually have diabetes.
- False Positive: The model predicts the patient has diabetes, but the patient does not actually have diabetes.

The confusion matrix is a tool used to assess the quality of a...

Legacy source slide 103; aligned new slide is recorded in slide_map.json.

- Confusion Matrix
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Suppose you have data for 100 patients. You create a model that...

Legacy source slide 104; aligned new slide is recorded in slide_map.json.

- Confusion Matrix Example
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Metrics

Legacy source slide 105; aligned new slide is recorded in slide_map.json.

- Accuracy: The number of correct predictions (true positives + true negatives) divided by the total number of predictions.
- Precision: The number of the cases classified as positive that are actually positive: the number of true positives divided by (the number of true positives plus false positives).
- Recall: The fraction of positive cases correctly identified: the number of true positives divided by (the number of true positives plus false negatives).
- F1 Score: An overall metric that essentially combines precision and recall.

Accuracy

Legacy source slide 106; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Precision

Legacy source slide 107; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Recall or Sensitivity

Legacy source slide 108; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Specificity

Legacy source slide 109; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

F1 Score

Legacy source slide 110; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Threshold

Legacy source slide 111; aligned new slide is recorded in slide_map.json.

- By default, a predicted probability including or above 0.5 results in a class prediction of 1, while a prediction below this threshold means that there's a greater probability…
- Designer has a useful threshold slider for reviewing how the model performance would change depending on the set threshold.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Threshold

Legacy source slide 112; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

ROC

Legacy source slide 113; aligned new slide is recorded in slide_map.json.

- Another term for recall is True positive rate, and it has a corresponding metric named False positive rate, which measures the number of negative cases incorrectly identified…
- Plotting these metrics against each other for every possible threshold value between 0 and 1 results in a curve, known as the ROC curve.

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- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

ROC

Legacy source slide 114; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

AUC

Legacy source slide 115; aligned new slide is recorded in slide_map.json.

- In an ideal model, the curve would go all the way up the left side and across the top, so that it covers the full area of the chart.
- The larger the area under the curve, of AUC metric, (which can be any value from 0 to 1), the better the model is performing.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Activity: Classification

Legacy source slide 116; aligned new slide is recorded in slide_map.json.

- Import the Iris flower dataset from internet https://raw.githubusercontent.com/pandas-dev/pandas/master/pandas/tests/data/iris.csv Create a Predictive Classification Model for…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Iris Flower Dataset

Legacy source slide 117; aligned new slide is recorded in slide_map.json.

- setosa (0)
- versicolor (1)
- virginica (2)
- Iris flower dataset, introduced in 1936 by Sir Ronald Fisher

Iris Flower Dataset

Legacy source slide 118; aligned new slide is recorded in slide_map.json.

- Features in the Iris dataset: sepal length in cm sepal width in cm petal length in cm petal width in cm Target classes to predict: setosa : 0 versicolor : 1 virginica : 2
- Mechanism: name the component, transformation and state transition.

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- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Hint for Activity

Legacy source slide 119; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Topic 3 — Computer vision

Clustering Example

Legacy source slide 121; aligned new slide is recorded in slide_map.json.

- Clustering groups unlabeled records by similarity; business meaning—not class accuracy—determines whether the groups are useful.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Create a Pipeline

Legacy source slide 122; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Add Data Transformations

Legacy source slide 123; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Select Columns

Legacy source slide 124; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.

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- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Clean Missing Data

Legacy source slide 125; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Normalize Data

Legacy source slide 126; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Run the Pipeline

Legacy source slide 127; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Add Training Modules

Legacy source slide 128; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Run the Training Pipeline

Legacy source slide 129; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.

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- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Evaluate a Clustering Model

Legacy source slide 130; aligned new slide is recorded in slide_map.json.

- Clustering groups unlabeled records by similarity; business meaning—not class accuracy—determines whether the groups are useful.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Create an Inference Pipeline

Legacy source slide 131; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Test the Service

Legacy source slide 132; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

What is Azure Automated Machine Learning?

Legacy source slide 133; aligned new slide is recorded in slide_map.json.

- Azure Machine Learning includes an automated machine learning capability that automatically tries multiple pre-processing techniques and model-training algorithms in parallel.
- These automated capabilities use the power of cloud compute to find the best performing supervised machine learning model for your data.
- Automated machine learning allows you to train models without extensive data science or programming knowledge For people with a data science and programming background, it…
- Control: define the threshold or failure that routes to human review.

Understand the AutoML process

Legacy source slide 134; aligned new slide is recorded in slide_map.json.

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- Prepare data: Identify the features and label in a dataset. Test the machine learning model for performance using the validation data set.
- Pre-process, or clean and transform, the data as needed. Evaluate performance: Compare how close the model's predictions are to the known labels.
- Train model: Split the data into two groups, a training and a validation set. Deploy a predictive service: After you train a machine learning model, you can deploy the model as…
- Train a machine learning model using the training data set.

Understand the AutoML process

Legacy source slide 135; aligned new slide is recorded in slide_map.json.

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- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure Machine Learning studio

Legacy source slide 136; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Prepare data

Legacy source slide 137; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Train model

Legacy source slide 138; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Configuration

Legacy source slide 139; aligned new slide is recorded in slide_map.json.

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- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Additional Configuration

Legacy source slide 140; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Additional Configuration

Legacy source slide 141; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Auto ML Experiment

Legacy source slide 142; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Auto ML Experiment

Legacy source slide 143; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Evaluate Performance

Legacy source slide 144; aligned new slide is recorded in slide_map.json.

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- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Review Result

Legacy source slide 145; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Model Summary

Legacy source slide 146; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Review the Best Model

Legacy source slide 147; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Deploy a Predictive Service

Legacy source slide 148; aligned new slide is recorded in slide_map.json.

- In Azure Machine Learning, you can deploy a service as an Azure Container Instances (ACI) or to an Azure Kubernetes Service (AKS) cluster.
- For production scenarios, an AKS deployment is recommended, for which you must create an inference cluster compute target.
- In this exercise, you'll use an ACI service, which is a suitable deployment target for testing, and does not require you to create an inference cluster.
- Control: define the threshold or failure that routes to human review.

Deploy a Model as a Service

Legacy source slide 149; aligned new slide is recorded in slide_map.json.

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- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Deploy a Model as a Service

Legacy source slide 150; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Topic 3 Explore Computer Vision

Legacy source slide 151; aligned new slide is recorded in slide_map.json.

- Pixels become model features; labels, regions or text must be compared with the source image.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

What is Computer Vision

Legacy source slide 152; aligned new slide is recorded in slide_map.json.

- Computer vision is an area of artificial intelligence (AI) in which software systems are designed to perceive the world visually, though cameras, images, and video.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Image Classification

Legacy source slide 153; aligned new slide is recorded in slide_map.json.

- A classifier maps a feature vector to a categorical label and exposes confidence or class probability for validation.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure Resources for Computer Vision

Legacy source slide 154; aligned new slide is recorded in slide_map.json.

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- To use the Computer Vision service, you need to create a resource for it in your Azure subscription.
- You can use either of the following resource types: Custom Vision: A dedicated resource for the custom vision service, which can be training, a prediction, or both resources.
- Cognitive Services: A general cognitive services resource that includes Custom Vision along with many other cognitive services.
- You can use this type of resource for training, prediction, or both

Model Evaluation

Legacy source slide 155; aligned new slide is recorded in slide_map.json.

- Precision: What percentage of the class predictions made by the model were correct? Average Precision (AP): An overall metric that takes into account both precision and recall
- For example, if the model predicted that 10 images are oranges, of which eight were actually oranges, then the precision is 0.8 (80%).
- Recall: What percentage of class predictions did the model correctly identify?
- For example, if there are 10 images of apples, and the model found 7 of them, then the recall is 0.7 (70%).

Using the model for prediction

Legacy source slide 156; aligned new slide is recorded in slide_map.json.

- Project ID: The unique ID of the Custom Vision project you created to train the model.
- Model name: The name you assigned to the model during publishing.
- Prediction endpoint: The HTTP address of the endpoints for the prediction resource to which you published the model (not the training resource).
- Prediction key: The authentication key for the prediction resource to which you published the model (not the training resource).

An object detection model might be used to identify the individual…

Legacy source slide 157; aligned new slide is recorded in slide_map.json.

- Object Detection
- The class of each object identified in the image.
- The probability score of the object classification (which you can interpret as the confidence of the predicted class being correct) The coordinates of a bounding box for each…
- Control: define the threshold or failure that routes to human review.

Face detection involves identifying regions of an image that contain…

Legacy source slide 158; aligned new slide is recorded in slide_map.json.

- Face Detection

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- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Moving beyond simple face detection, some algorithms can also return...

Legacy source slide 159; aligned new slide is recorded in slide_map.json.

- Face Analysis
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

A further application of facial analysis is to train a machine...

Legacy source slide 160; aligned new slide is recorded in slide_map.json.

- Face Recognition
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Face Analysis in Azure

Legacy source slide 161; aligned new slide is recorded in slide_map.json.

- Computer Vision, which offers face detection and some basic face analysis, such as returning the bounding box coordinates around an image.
- Video Indexer, which you can use to detect and identify faces in a video.
- Face, which offers pre-built algorithms that can detect, recognize, and analyze faces.
- Control: define the threshold or failure that routes to human review.

Open 01d - Face Analysis.ipynb Run the code as per jupyter file

Legacy source slide 162; aligned new slide is recorded in slide_map.json.

- Activity: Face Analysis
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Read API

Legacy source slide 163; aligned new slide is recorded in slide_map.json.

- The Read API uses the latest recognition models and is optimized for images that have a significant amount of text or has considerable visual noise.
- The Read API can handle scanned documents that have a lot of text.
- It also has the ability to automatically determine the proper recognition model to use, taking into consideration lines of text and supporting images with printed text as well…

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- Control: define the threshold or failure that routes to human review.

Open 01e - Optical Character Recognition.ipynb Run the code as per…

Legacy source slide 164; aligned new slide is recorded in slide_map.json.

- Activity: OCR
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Azure AI Document Intelligence

Legacy source slide 165; aligned new slide is recorded in slide_map.json.

- Azure AI Document Intelligence supports automated document processing through: A pre-built receipt model that is provided out-of-the-box, and is trained to recognize and…
- Custom models are trained using your own data, which helps to tailor this model to your specific forms.
- Starting with only five samples of your forms, you can train the custom model.
- Control: define the threshold or failure that routes to human review.

Pre-built Receipt Model

Legacy source slide 166; aligned new slide is recorded in slide_map.json.

- The model is able to extract key information from the receipt slip: time of transaction date of transaction merchant information taxes paid receipt totals other pertinent…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of Computer Vision

Legacy source slide 167; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/analyze-images-computer-vision/5-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.explore-computer-v…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of Facial Recognition

Legacy source slide 168; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/detect-analyze-faces/5-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.explore-computer-vision-micr…

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- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Topic 4 — Natural language processing

Topic 4 Explore Natural Language Processing

Legacy source slide 170; aligned new slide is recorded in slide_map.json.

- Normalized text produces typed outputs such as language, sentiment, entities or redacted spans.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Natural Language Processing

Legacy source slide 171; aligned new slide is recorded in slide_map.json.

- Natural language processing supports applications that can see, hear, speak with, and understand users.
- Text analytics is a process where an artificial intelligence (AI) algorithm, running on a computer, evaluates these same attributes in text, to determine specific insights
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Use the language detection capability of the Language service to...

Legacy source slide 172; aligned new slide is recorded in slide_map.json.

- Language Detection
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Sentiment Analysis

Legacy source slide 173; aligned new slide is recorded in slide_map.json.

- The text analytics capabilities in the Language service can evaluate text and return sentiment scores and labels for each sentence.
- This capability is useful for detecting positive and negative sentiment in social media, customer reviews, discussion forums and more.
- Using the pre-built machine learning classification model, the service evaluates the text and returns a sentiment score in the range of 0 to 1, with values closer to 1 being a…
- Scores that are close to the middle of the range (0.5) are considered neutral or indeterminate.

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Key Phrase Extraction

Legacy source slide 174; aligned new slide is recorded in slide_map.json.

- table friendly hostess dinner ambiance place
- attentive service great food birthday celebration fantastic experience
- Key phrase extraction is the concept of evaluating the text of a document, or documents, and then identifying the main talking points of the document(s).
- Key phrase extraction can provide some context to this review by extracting the following phrases:

You can provide the Language service with unstructured text and it...

Legacy source slide 175; aligned new slide is recorded in slide_map.json.

- Entity Recognition
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Open 02a - Text Analytics.ipynb Run the code as per jupyter file...

Legacy source slide 176; aligned new slide is recorded in slide_map.json.

- Activity: Text Analytics
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Speech Recognition

Legacy source slide 177; aligned new slide is recorded in slide_map.json.

- Speech recognition is concerned with taking the spoken word and converting it into data that can be processed - often by transcribing it into a text representation. A language…
- The spoken words can be in the form of a recorded voice in an audio file, or live audio from a microphone.
- Speech patterns are analyzed in the audio to determine recognizable patterns that are mapped to words.
- To accomplish this feat, the software typically uses multiple types of models, including: An acoustic model that converts the audio signal into phonemes (representations of…

Speech Synthesis

Legacy source slide 178; aligned new slide is recorded in slide_map.json.

- Speech synthesis is in many respects the reverse of speech recognition. To synthesize speech, the system typically tokenizes the text to break it down into individual words,…

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- It is concerned with vocalizing data, usually by converting text to speech. It then breaks the phonetic transcription into prosodic units (such as phrases, clauses, or…
- A speech synthesis solution typically requires the following information: The text to be spoken.
- The voice to be used to vocalize the speech.

Speech Service

Legacy source slide 179; aligned new slide is recorded in slide_map.json.

- The Speech service includes the following application programming interfaces (APIs): Speech-to-text - used to transcribe speech from an audio source to text format.
- Text-to-speech - used to generate spoken audio from a text source.
- Speech Translation - used to translate speech in one language to text or speech in another.
- Control: define the threshold or failure that routes to human review.

Speech-to-Text API

Legacy source slide 180; aligned new slide is recorded in slide_map.json.

- You can use the speech-to-text API to perform real-time or batch transcription of audio into a text format. The model is optimized for two scenarios, conversational and dictation.
- The audio source for transcription can be a real-time audio stream from a microphone or an audio file. You can also create and train your own custom models including acoustics,…
- The model that is used by the speech-to-text API, is based on the Universal Language Model that was trained by Microsoft.
- The data for the model is Microsoft-owned and deployed to Microsoft Azure.

Text-to-Speech API

Legacy source slide 181; aligned new slide is recorded in slide_map.json.

- The text-to-speech API enables you to convert text input to audible speech, which can either be played directly through a computer speaker or written to an audio file.
- When you use the text-to-speech API, you can specify the voice to be used to vocalize the text.
- This capability offers you the flexibility to personalize your speech synthesis solution and give it a specific character.
- Control: define the threshold or failure that routes to human review.

Open 02b - Speech.ipynb Run the code as per jupyter file Turn the…

Legacy source slide 182; aligned new slide is recorded in slide_map.json.

- Activity: Speech
- response_text = 'Turning the light on.'

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- response_text = 'Turning the light off.'
- Control: define the threshold or failure that routes to human review.

Text and Speech Translation

Legacy source slide 183; aligned new slide is recorded in slide_map.json.

- Text translation can be used to translate documents from one language to another, translate email communications that come from foreign governments, and even provide the…
- Many times you will see a Translate option for posts on social media sites, or the Bing search engine can offer to translate entire web pages that are turned in search results.
- Speech translation is used to translate between spoken languages, sometimes directly (speech-to-speech translation) and sometimes by translating to an intermediary text format…
- Control: define the threshold or failure that routes to human review.

Open 02d - Language Understanding.ipynb Run the code as per jupyter…

Legacy source slide 184; aligned new slide is recorded in slide_map.json.

- Activity: Text and Speech Translation
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Conversational language understanding

Legacy source slide 185; aligned new slide is recorded in slide_map.json.

- Azure has a Language Understanding web application (https://www.Microsoft Foundry / Azure AI Language/) that provides machine learning-based service to build natural language…
- You can quickly create enterprise-ready, custom models that continuously improve.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Open 02d - Language Understanding.ipynb Run the code as per jupyter…

Legacy source slide 186; aligned new slide is recorded in slide_map.json.

- Activity: Language Understanding
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Think about how often you communicate with other people through…

Legacy source slide 187; aligned new slide is recorded in slide_map.json.

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- Conversational AI
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Conversational AI

Legacy source slide 188; aligned new slide is recorded in slide_map.json.

- Bots are designed to interact with users in a conversational manner, as shown in this example of a chat interface:
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Custom question answering and Azure Bot Service

Legacy source slide 189; aligned new slide is recorded in slide_map.json.

- You can easily create a user support bot solution on Microsoft Azure using a combination of two core technologies: Custom question answering.
- This cognitive service enables you to create and publish a knowledge base with built-in natural language processing capabilities.
- Azure Bot Service.
- This service provides a framework for developing, publishing, and managing bots on Azure.

Azure Custom question answering

Legacy source slide 190; aligned new slide is recorded in slide_map.json.

- Azure has a Language Understanding web application (https://www.Microsoft Foundry / Azure AI Language/) allow one to build, train and publish a sophisticated bot using FAQ…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Open Run the code as per jupyter file You will need to create an app...

Legacy source slide 191; aligned new slide is recorded in slide_map.json.

- Activity: Custom question answering
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Define questions and answers

Legacy source slide 192; aligned new slide is recorded in slide_map.json.

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- After provisioning a Language service resource, you can use the Language Studio's custom question answering feature to create a knowledge base that consists of…
- These questions and answers can be: Generated from an existing FAQ document or web page.
- Entered and edited manually.
- Control: define the threshold or failure that routes to human review.

Test the Knowledge Base

Legacy source slide 193; aligned new slide is recorded in slide_map.json.

- After creating a set of question-and-answer pairs, you must save it.
- This process analyzes your literal questions and answers and applies a built-in natural language processing model to match appropriate answers to questions, even when they are…
- Then you can use the built-in test interface in the Language Studio to test your knowledge base by submitting questions and reviewing the answers that are returned.
- Control: define the threshold or failure that routes to human review.

Use the knowledge base

Legacy source slide 194; aligned new slide is recorded in slide_map.json.

- When you're satisfied with your knowledge base, deploy it.
- Then you can use it over its REST interface.
- To access the knowledge base, client applications require: The knowledge base ID The knowledge base endpoint The knowledge base authorization key
- Control: define the threshold or failure that routes to human review.

Open 03a - QnA Bot.ipynb Follow the instructions on the notebook You...

Legacy source slide 195; aligned new slide is recorded in slide_map.json.

- Activity: Custom question answering
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Build a bot with the Azure Bot Service

Legacy source slide 196; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

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Conversational Language Understanding

Legacy source slide 197; aligned new slide is recorded in slide_map.json.

- On Microsoft Azure, conversational language understanding is supported through the Language Service.
- To work with Conversational Language Understanding, you need to take into account three core concepts: utterances entities intents.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

An utterance is an example of something a user might say, and which...

Legacy source slide 198; aligned new slide is recorded in slide_map.json.

- Utterances
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

An entity is an item to which an utterance refers. For example, fan...

Legacy source slide 199; aligned new slide is recorded in slide_map.json.

- Entities
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

An intent represents the purpose, or goal, expressed in a user's...

Legacy source slide 200; aligned new slide is recorded in slide_map.json.

- Intents
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of Text Analysis with the Language Service

Legacy source slide 201; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/analyze-text-with-text-analytics-service/5-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.explore-…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of question answering with the...

Legacy source slide 202; aligned new slide is recorded in slide_map.json.

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- https://learn.microsoft.com/en-us/training/modules/build-faq-chatbot-qna-maker-azure-bot-service/5-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.exp…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of conversational language understanding

Legacy source slide 203; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/create-language-model-with-language-understanding/5-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of Azure AI Speech

Legacy source slide 204; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/recognize-synthesize-speech/5-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.explore-natural-langu…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of language translation

Legacy source slide 205; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/translate-text-with-translation-service/3a-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.explore-…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Topic 5 — Decision support and anomaly evidence

How Anomaly Detector works

Legacy source slide 207; aligned new slide is recorded in slide_map.json.

- The Anomaly Detector service identifies anomalies that exist outside the scope of a boundary. If a value exceeds either boundary, it will be identified as an anomaly.

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- The boundary is set using a sensitivity value. You can adjust the boundaries by applying a marginScale to the upper and lower margins as demonstrated by the following formula.…
- By default, the upper and lower boundaries for anomaly detection are calculated using concepts known as expectedValue, upperMargin, and lowerMargin.
- The upper and lower boundaries are calculated using these three values.

Data format

Legacy source slide 208; aligned new slide is recorded in slide_map.json.

- Inspect the named input, processing boundary, output artifact and acceptance evidence before approving the service.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

When to use Anomaly Detector

Legacy source slide 209; aligned new slide is recorded in slide_map.json.

- The Anomaly Detector service supports batch processing of time series data and last-point anomaly detection for real-time data.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Batch Detection

Legacy source slide 210; aligned new slide is recorded in slide_map.json.

- Consider a pharmaceutical company that stores medications in storage facilities where the temperature in the facilities needs to remain within a specific range.
- To evaluate whether the medication remained stored in a safe temperature range in the past three months we need to know: the maximum allowable temperature the minimum allowable…
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Batch Detection Example

Legacy source slide 211; aligned new slide is recorded in slide_map.json.

- Batch detection involves applying the algorithm to an entire data series at one time. Batch detection is best used when your data contains: Flat trend time series data with…
- The concept of time series data involves evaluation of a data set as a batch. Examples would be hourly, daily, or monthly patterns.

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- Use your time series to detect any anomalies that might exist throughout your data. Using seasonal data, and specifying a period for that pattern, can help to reduce the…
- This operation generates a model using your entire time series data, with each point analyzed using the same model.

Real-time Detection

Legacy source slide 212; aligned new slide is recorded in slide_map.json.

- Real-time detection uses streaming data by comparing previously seen data points to the last data point to determine if your latest one is an anomaly.
- This operation generates a model using the data points you send, and determines if the target (current) point is an anomaly.
- By calling the service with each new data point you generate, you can monitor your data as it's created.
- Control: define the threshold or failure that routes to human review.

Real-time Detection Example

Legacy source slide 213; aligned new slide is recorded in slide_map.json.

- Consider a scenario in the carbonated beverage industry where real-time anomaly detection may be useful. If the temperature is too warm, the carbon dioxide will not adhere…
- The carbon dioxide added to soft drinks during the bottling or canning process needs to stay in a specific temperature range.
- Bottling systems use a device known as a carbo-cooler to achieve the refrigeration of the product for this process.
- If the temperature goes too low, the product will freeze in the carbo-cooler.

Topic 6 — Knowledge mining

Azure AI Search

Legacy source slide 215; aligned new slide is recorded in slide_map.json.

- Azure AI Search provides the infrastructure and tools to create search solutions that extract data from various structured, semi-structured, and non-structured documents
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Understand Indexes

Legacy source slide 216; aligned new slide is recorded in slide_map.json.

- An Azure AI Search index can be thought of as a container of searchable documents.

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- Conceptually you can think of an index as a table and each row in the table represents a document.
- Tables have columns, and the columns can be thought of as equivalent to the fields in a document.
- Columns have data types, just as the fields do on the documents.

Index Schema

Legacy source slide 217; aligned new slide is recorded in slide_map.json.

- The search contract joins a key field, searchable content, filters and an ingestion path that emits indexed JSON documents.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Index Attributes

Legacy source slide 218; aligned new slide is recorded in slide_map.json.

- The search contract joins a key field, searchable content, filters and an ingestion path that emits indexed JSON documents.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Use an indexer to build an index

Legacy source slide 219; aligned new slide is recorded in slide_map.json.

- Azure AI Search lets you create and load JSON documents into an index with two approaches: Push method: JSON data is pushed into a search index via either the REST API or the…
- Pushing data has the most flexibility as it has no restrictions on the data source type, location, or frequency of execution.
- Pull method: Search service indexers can pull data from popular Azure data sources, and if necessary, export that data into JSON if it isn't already in that format.
- Control: define the threshold or failure that routes to human review.

Knowledge Store

Legacy source slide 220; aligned new slide is recorded in slide_map.json.

- A knowledge store is persistent storage of enriched content.
- The purpose of a knowledge store is to store the data generated from AI enrichment in a container.
- For example, you may want to save the results of an AI skillset that generates captions from images.
- Control: define the threshold or failure that routes to human review.

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Create an index in the Azure portal

Legacy source slide 221; aligned new slide is recorded in slide_map.json.

- The search contract joins a key field, searchable content, filters and an ingestion path that emits indexed JSON documents.
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Query data in an Azure AI Search index

Legacy source slide 222; aligned new slide is recorded in slide_map.json.

- Index and query design are closely linked. Queries can specify what fields are searched and returned, how search results are shaped, and how the results should be filtered or…
- After we build the index, we can perform queries. A query that doesn't specify the field to search will execute against all the searchable fields within the index
- A crucial component to understand is that the schema of the index determines what queries can be answered.
- Azure AI Search queries can be submitted as an HTTP or REST API request, with the response coming back as JSON.

Simple query requests

Legacy source slide 223; aligned new slide is recorded in slide_map.json.

- A query request is a list or words (search terms) and query operators (simple or full) of what you would like to see returned in a result set.
- Let's look what components make up a search query.
- Consider this simple search example coffee (-"busy" + "wifi")
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of Azure AI Document Intelligence

Legacy source slide 224; aligned new slide is recorded in slide_map.json.

- https://learn.microsoft.com/en-us/training/modules/analyze-receipts-form-recognizer/5-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.document-intelli…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Knowledge Quiz - Fundamentals of Knowledge Mining and Azure AI Search

Legacy source slide 225; aligned new slide is recorded in slide_map.json.

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- https://learn.microsoft.com/en-us/training/modules/intro-to-azure-search/8-knowledge-check?ns-enrollment-type=learningpath&ns-enrollment-id=learn.wwl.document-intelligence-knowl…
- Mechanism: name the component, transformation and state transition.
- Output evidence: record the schema, metric, confidence or observable result.
- Control: define the threshold or failure that routes to human review.

Hands-on labs

The following procedures are mirrored from the individual lab folders. If a portal label changes, follow the named resource/state and record the difference in the issue log.

Lab 01 — AI Workloads, Responsible AI, and Microsoft Foundry

Duration: 55 minutes · Alignment: K1, K6, K7, A1

Lab folder: labs/lab-01-ai-workloads-responsible-ai-foundry/README.md

Classify AI workloads, map them to Azure services, inspect a Foundry project, and record responsible-AI controls.

Scenario: Contoso Retail needs a defensible service map before its implementation partner provisions any resource.

Deliverable: A workload-to-service decision map, responsible-AI control register, and Foundry project evidence pack.

Detailed procedure

7. Open the Microsoft Foundry portal and sign in with the Azure account provided for training.

   Evidence checkpoint: mark item 1 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

8. Create or select a resource group dedicated to the class; record subscription, region, resource-group name and owner.

   Evidence checkpoint: mark item 2 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

9. Create a Foundry project with a unique name and capture the project overview showing project and parent resource.

   Evidence checkpoint: mark item 3 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

10. Inspect the project endpoint and connected resources; redact keys or tokens from all screenshots.

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Evidence checkpoint: mark item 4 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

11. Classify the six scenario cards in samples/workloads.csv by input, desired output, AI workload and suitable Azure service.

Evidence checkpoint: mark item 5 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

12. For each mapping, name the human decision owner and the evidence that would prove the output is acceptable.

Evidence checkpoint: mark item 6 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

13. Complete evidence/responsible-ai-checklist.md for fairness, reliability and safety, privacy and security, inclusiveness, transparency and accountability.

Evidence checkpoint: mark item 7 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

14. Record one limitation or uncertainty in evidence/issue-log.md and assign severity, owner and next action.

Evidence checkpoint: mark item 8 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

15. Validate that no secret appears in the evidence pack and that every workload has a service and control.

Evidence checkpoint: mark item 9 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

16. If you created billable resources, delete the lab resource group after the trainer verifies your evidence.

Evidence checkpoint: mark item 10 in labs/lab-01-ai-workloads-responsible-ai-foundry/evidence/checklist.md.

Acceptance checks

- [ ] Six workload rows classified
- [ ] Six responsible-AI principles addressed
- [ ] Project/resource boundary visible
- [ ] No secret values captured
- [ ] One issue assigned to an owner

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.

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- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

Lab 02 — Machine Learning Fundamentals and Metrics

Duration: 55 minutes · Alignment: K2, K3, K4

Lab folder: labs/lab-02-machine-learning-fundamentals-metrics/README.md

Differentiate regression, classification and clustering and compute evidence-based model metrics.

Scenario: A bike-share team must choose the correct technique and reject a misleading model before deployment.

Deliverable: A technique-selection table and verified MAE, RMSE, precision, recall and confusion-matrix calculations.

Detailed procedure

17. Open samples/scenarios.csv and label each case regression, classification or clustering; justify the predicted output type.

    Evidence checkpoint: mark item 1 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

18. Identify features and label for the demand-forecasting and churn examples; flag any field that leaks the answer.

    Evidence checkpoint: mark item 2 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

19. Split the sample rows into training and validation partitions using the supplied split column.

    Evidence checkpoint: mark item 3 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

20. Use samples/model-results.csv to calculate absolute error for every regression row.

    Evidence checkpoint: mark item 4 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

21. Calculate MAE and RMSE; explain why RMSE increases the influence of large errors.

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Evidence checkpoint: mark item 5 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

22. Build a 2x2 confusion matrix from actual_class and predicted_class.

Evidence checkpoint: mark item 6 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

23. Calculate precision and recall for the positive class and compare them with accuracy.

Evidence checkpoint: mark item 7 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

24. Inspect the clustering sample and explain why it has no label during training.

Evidence checkpoint: mark item 8 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

25. Document one data-quality defect, its likely metric impact, and a correction.

Evidence checkpoint: mark item 9 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

26. Save completed calculations and the technique-selection table under evidence/.

Evidence checkpoint: mark item 10 in labs/lab-02-machine-learning-fundamentals-metrics/evidence/checklist.md.

Acceptance checks

- [ ] Every scenario has a justified technique
- [ ] MAE and RMSE are reproducible
- [ ] Confusion matrix totals match source rows
- [ ] Precision and recall show working
- [ ] Leakage risk documented

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.
- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

Lab 03 — Azure Machine Learning AutoML and Model Lifecycle

Duration: 70 minutes · Alignment: K3, K4, K6, A1, A2

Lab folder: labs/lab-03-azure-machine-learning-automl-lifecycle/README.md

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Configure an AutoML regression experiment, interpret ranking evidence, and document deployment controls.

Scenario: Contoso Mobility needs a repeatable demand-forecast experiment that its service provider can review.

Deliverable: An AutoML experiment record with data, task, target, compute, metric, best-model evidence and deployment decision.

Detailed procedure

27. Create or open an Azure Machine Learning workspace in the trainer-approved subscription and region.

    Evidence checkpoint: mark item 1 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

28. Create a tabular data asset from samples/daily-bike-share.csv and preview the schema.

    Evidence checkpoint: mark item 2 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

29. Confirm rentals is the numeric target and exclude date-derived identifiers that could leak the answer.

    Evidence checkpoint: mark item 3 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

30. Start an Automated ML regression job, select serverless or trainer-approved compute, and set a conservative timeout.

    Evidence checkpoint: mark item 4 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

31. Choose normalized root mean squared error or another trainer-approved primary metric and record the choice.

    Evidence checkpoint: mark item 5 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

32. Submit the job and capture the job status, experiment name and data asset version.

    Evidence checkpoint: mark item 6 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

33. When complete, inspect the model leaderboard, validation metric and residual or predicted-versus-true chart.

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Evidence checkpoint: mark item 7 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

34. Open the best model and record algorithm, transformations and feature-importance evidence where available.

Evidence checkpoint: mark item 8 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

35. Complete evidence/model-card.md with intended use, limitations, metric threshold, owner and monitoring trigger.

Evidence checkpoint: mark item 9 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

36. Do not deploy unless instructed; document the endpoint authentication and monitoring controls that would be required.

Evidence checkpoint: mark item 10 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

37. Delete compute or the resource group after evidence review to avoid unnecessary cost.

Evidence checkpoint: mark item 11 in labs/lab-03-azure-machine-learning-automl-lifecycle/evidence/checklist.md.

Acceptance checks

- [ ] Versioned data asset recorded
- [ ] Regression target and metric justified
- [ ] Best model evidence captured
- [ ] Model card includes limitations
- [ ] Cleanup confirmed

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.
- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

Lab 04 — Computer Vision, Image Analysis, and OCR

Duration: 65 minutes · Alignment: K3, K7, K8, A2, A3

Lab folder: labs/lab-04-computer-vision-image-analysis-ocr/README.md

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Analyze an image, compare vision outputs with source evidence, and report accuracy or privacy issues.

Scenario: A product-quality team needs visual inspection and text extraction without treating model output as ground truth.

Deliverable: An image-analysis result, OCR extraction, confidence review and issue report.

Detailed procedure

38. Open a Foundry project and deploy or select a trainer-approved vision-capable model.

    Evidence checkpoint: mark item 1 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

39. Upload samples/store-shelf.jpg and ask for objects, scene description and any visible text.

    Evidence checkpoint: mark item 2 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

40. Record the prompt, model/deployment name and response in evidence/vision-results.md.

    Evidence checkpoint: mark item 3 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

41. Compare each detected object with the image; mark true positive, false positive or missed object.

    Evidence checkpoint: mark item 4 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

42. Open the Content Understanding or information-extraction playground and select the Read/OCR analyzer.

    Evidence checkpoint: mark item 5 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

43. Upload samples/receipt.png and run analysis.

    Evidence checkpoint: mark item 6 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

44. Compare extracted text and bounding regions with the image; record at least two correct and one uncertain item.

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Evidence checkpoint: mark item 7 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

45. State the confidence or business rule that triggers human review when confidence is unavailable or low.

Evidence checkpoint: mark item 8 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

46. Complete evidence/issue-log.md with reproduction steps, expected result, actual result, impact and owner.

Evidence checkpoint: mark item 9 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

47. Redact personal information from evidence and clean up billable resources.

Evidence checkpoint: mark item 10 in labs/lab-04-computer-vision-image-analysis-ocr/evidence/checklist.md.

Acceptance checks

- [ ] Image input and model recorded
- [ ] Output compared with source image
- [ ] OCR uncertainty identified
- [ ] Issue report is reproducible
- [ ] Privacy and cleanup controls applied

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.
- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

Lab 05 — Language: Text Analysis, PII, and Clean Inputs

Duration: 60 minutes · Alignment: K5, K7, K9, K10, K11, A3, A5

Lab folder: labs/lab-05-language-text-analysis-pii/README.md

Analyze text, detect language and PII, clean malformed input, and verify the output contract.

Scenario: A service desk wants useful themes from feedback while preventing personal data from entering reports.

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Deliverable: A before/after clean-data record, text-analysis output, PII-redaction evidence and schema validation.

Detailed procedure

48. Open samples/customer-feedback.csv and identify blank, duplicate, mixed-language and PII-bearing rows.

    Evidence checkpoint: mark item 1 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

49. Copy the file to evidence/customer-feedback-clean.csv and remove duplicates while preserving one auditable row identifier.

    Evidence checkpoint: mark item 2 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

50. Normalize whitespace and blank values; do not replace unknown facts with invented values.

    Evidence checkpoint: mark item 3 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

51. Use a general-purpose model to summarize the cleaned feedback into three evidence-backed themes.

    Evidence checkpoint: mark item 4 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

52. Use the specialized text-analysis tool to detect language for the supplied multilingual samples.

    Evidence checkpoint: mark item 5 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

53. Run PII detection and record category, span and redacted output without exposing the original PII in screenshots.

    Evidence checkpoint: mark item 6 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

54. Inspect the generated client-code example and identify endpoint, credential mechanism, request payload and response fields.

    Evidence checkpoint: mark item 7 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

55. Validate evidence/text-analysis.json against samples/text-analysis-schema.json.

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Evidence checkpoint: mark item 8 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

56. Record any NaN/unknown language, over-redaction or missed PII as an issue with severity and owner.

Evidence checkpoint: mark item 9 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

57. Delete resources or the lab resource group after trainer verification.

Evidence checkpoint: mark item 10 in labs/lab-05-language-text-analysis-pii/evidence/checklist.md.

Acceptance checks

- [ ] Clean dataset preserves row IDs
- [ ] Summary cites source themes
- [ ] Language and PII outputs recorded
- [ ] JSON validates against schema
- [ ] No raw PII is published

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.
- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

Lab 06 — Speech, Translation, and Conversational AI

Duration: 60 minutes · Alignment: K7, K8, K9, A2, A3

Lab folder: labs/lab-06-speech-translation-conversational-ai/README.md

Use speech input/output, design a translation contract, and define a human handoff for conversational AI.

Scenario: A multilingual support team needs a voice interaction that fails safely in noisy conditions.

Deliverable: A transcript/output evidence pack, translation message contract and escalation decision table.

Detailed procedure

58. Create or open the Foundry project and select a trainer-approved speech-capable agent or Speech resource.

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Evidence checkpoint: mark item 1 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

59. Configure a neutral system instruction that states the support scope and prohibited actions.

Evidence checkpoint: mark item 2 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

60. Connect the Speech Voice Live experience or trainer-approved Speech interface.

Evidence checkpoint: mark item 3 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

61. Read samples/support-utterances.txt in a quiet setting and record transcription evidence.

Evidence checkpoint: mark item 4 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

62. Repeat one utterance with background noise or a deliberate accent variation and compare the transcript.

Evidence checkpoint: mark item 5 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

63. Generate speech from the approved response and note voice, language and output format.

Evidence checkpoint: mark item 6 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

64. Complete samples/translation-contract.json with source language, target language, content and correlation ID.

Evidence checkpoint: mark item 7 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

65. Design a handoff rule using confidence/uncertainty, harmful-content flag and repeated-failure count.

Evidence checkpoint: mark item 8 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

66. Record consent, retention and accessibility controls in evidence/conversation-controls.md.

Evidence checkpoint: mark item 9 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

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67. Remove recordings containing personal data and clean up billable resources.

Evidence checkpoint: mark item 10 in labs/lab-06-speech-translation-conversational-ai/evidence/checklist.md.

Acceptance checks

- [ ] Speech input and output both evidenced
- [ ] Noise/accent difference discussed
- [ ] Translation contract is valid JSON
- [ ] Handoff rule is testable
- [ ] Consent and retention addressed

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.
- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

Lab 07 — Document Intelligence and Knowledge Mining

Duration: 70 minutes · Alignment: K5, K8, K9, K10, K11, A4, A5

Lab folder: labs/lab-07-document-intelligence-knowledge-mining/README.md

Extract document fields, normalize them to a data contract, and prepare searchable knowledge records.

Scenario: Accounts payable needs receipt data that can move safely from extraction into search and reporting systems.

Deliverable: A receipt extraction, cleaned interoperable JSON document, field-quality report and search-index design.

Detailed procedure

68. Open the Content Understanding or information-extraction playground in the Foundry project.

Evidence checkpoint: mark item 1 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

69. Run the Read analyzer on samples/receipt.png and capture raw OCR evidence.

Evidence checkpoint: mark item 2 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

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70. Run the receipt or layout analyzer and compare structured fields with the visible document.

Evidence checkpoint: mark item 3 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

71. Record merchant, date, currency, subtotal, tax and total with confidence or uncertainty notes.

Evidence checkpoint: mark item 4 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

72. Transform the result into evidence/receipt-normalized.json using the supplied canonical field names and ISO date/currency conventions.

Evidence checkpoint: mark item 5 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

73. Validate the normalized document against samples/receipt-schema.json and correct type/required-field failures.

Evidence checkpoint: mark item 6 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

74. Design an Azure AI Search index with key, searchable, filterable, sortable and facetable attributes.

Evidence checkpoint: mark item 7 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

75. Map the normalized document fields into the index and identify which fields must not be searchable.

Evidence checkpoint: mark item 8 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

76. Define a human-review queue for missing totals, low confidence and arithmetic mismatch.

Evidence checkpoint: mark item 9 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

77. Document RBAC, data retention, lineage and cleanup controls.

Evidence checkpoint: mark item 10 in labs/lab-07-document-intelligence-knowledge-mining/evidence/checklist.md.

Acceptance checks

- [ ] Raw and structured outputs compared

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- [ ] Normalized JSON validates
- [ ] Index attributes are justified
- [ ] Low-quality records route to review
- [ ] Lineage and access controls documented

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.
- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

Lab 08 — Generative AI and Agents in Microsoft Foundry

Duration: 65 minutes · Alignment: K1, K6, K7, K8, A1, A2, A3

Lab folder: labs/lab-08-generative-ai-foundry-agent/README.md

Deploy a model, configure instructions and grounding, evaluate responses, and record safety controls.

Scenario: A policy assistant must answer only from approved material and escalate when evidence is absent.

Deliverable: A model deployment and agent configuration with grounded tests, refusal evidence and model-card controls.

Detailed procedure

78. Create or open a Foundry project and inspect model catalog availability for the trainer-approved region.

    Evidence checkpoint: mark item 1 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

79. Deploy a small trainer-approved chat model; record deployment name, model version and quota/cost note.

    Evidence checkpoint: mark item 2 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

80. Run the baseline prompt in samples/prompt-tests.csv and capture the response.

    Evidence checkpoint: mark item 3 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

81. Add system instructions that constrain scope, require source attribution and specify an uncertainty response.

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Evidence checkpoint: mark item 4 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

82. Attach the supplied policy document or trainer-approved knowledge source for grounding.

Evidence checkpoint: mark item 5 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

83. Save the configuration as an agent if the service is available; otherwise preserve the playground configuration evidence.

Evidence checkpoint: mark item 6 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

84. Run grounded, out-of-scope, ambiguous and unsafe test cases from samples/prompt-tests.csv.

Evidence checkpoint: mark item 7 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

85. Score groundedness, relevance, completeness and safety in evidence/evaluation.csv.

Evidence checkpoint: mark item 8 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

86. Inspect the generated client-code example and identify endpoint, authentication and message payload fields.

Evidence checkpoint: mark item 9 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

87. Complete evidence/model-card.md and delete the deployment/resource group when instructed.

Evidence checkpoint: mark item 10 in labs/lab-08-generative-ai-foundry-agent/evidence/checklist.md.

Acceptance checks

- [ ] Model version and deployment recorded
- [ ] Instructions constrain scope
- [ ] Grounded and refusal tests captured
- [ ] Evaluation uses explicit criteria
- [ ] Cost and cleanup documented

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.

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- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

Lab 09 — Anomaly Evidence, Content Safety, Security, and Governance

Duration: 60 minutes · Alignment: K6, K7, K8, A3

Lab folder: labs/lab-09-anomaly-safety-governance/README.md

Compute anomaly evidence, test content-safety categories, and produce an actionable issue/governance report.

Scenario: Operations must replace a retired managed anomaly service without losing detection evidence or escalation controls.

Deliverable: A reproducible anomaly table, safety-severity results, issue report, access model and cost/cleanup checklist.

Detailed procedure

88. Open samples/timeseries.csv and calculate the rolling mean and residual columns using the supplied formula notes.

    Evidence checkpoint: mark item 1 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

89. Apply the stated threshold rule and mark each point normal or anomalous; do not tune the threshold after seeing the answer.

    Evidence checkpoint: mark item 2 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

90. Chart observed value, baseline and alert threshold; annotate the first true alert and one false-positive risk.

    Evidence checkpoint: mark item 3 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

91. Document why Azure AI Anomaly Detector retirement requires a migration plan rather than continued provisioning instructions.

    Evidence checkpoint: mark item 4 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

92. Open the Foundry safety/guardrails experience or the trainer-approved Content Safety interface.

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Evidence checkpoint: mark item 5 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

93. Run the supplied safe, borderline and unsafe samples; record category and severity without adding harmful detail.

Evidence checkpoint: mark item 6 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

94. Compare the service output with the organisation's escalation threshold and record disagreements.

Evidence checkpoint: mark item 7 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

95. Complete evidence/issue-report.md with reproduction, expected/actual results, evidence, impact, owner and remediation.

Evidence checkpoint: mark item 8 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

96. Add RBAC, key protection, logging, budget and human-review controls to evidence/governance-checklist.md.

Evidence checkpoint: mark item 9 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

97. Clean up resources and retain only redacted evidence.

Evidence checkpoint: mark item 10 in labs/lab-09-anomaly-safety-governance/evidence/checklist.md.

Acceptance checks

- [ ] Anomaly rule is reproducible
- [ ] Retirement/migration risk stated
- [ ] Safety categories and severities recorded
- [ ] Issue report has owner and remediation
- [ ] Security, cost and cleanup controls complete

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.
- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

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Lab 10 — AI-900 Capstone: End-to-End Azure AI Solution Review

Duration: 80 minutes · Alignment: K1-K11, A1-A5

Lab folder: labs/lab-10-ai900-capstone-course-review/README.md

Integrate service selection, workflow deployment, issue reporting, interoperability, cleaning and governance into one evidence-backed design.

Scenario: Contoso must approve one cross-service Azure AI solution for product feedback, receipts and policy support.

Deliverable: A final architecture, data contract, evidence matrix, issue register, validation plan and resource-cleanup record.

Detailed procedure

98. Choose one supplied capstone scenario and list its users, inputs, required outputs, data classification and success measures.

Evidence checkpoint: mark item 1 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

99. Select Azure services for ingestion, AI processing, knowledge retrieval and user interaction; justify each boundary.

Evidence checkpoint: mark item 2 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

100. Draw the end-to-end flow using the supplied architecture template and label every persisted data store.

Evidence checkpoint: mark item 3 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

101. Define the canonical JSON data contract and map source fields to target fields.

Evidence checkpoint: mark item 4 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

102. Apply the cleaning rules for missing, duplicate, malformed and sensitive data.

Evidence checkpoint: mark item 5 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

103. Link at least one evidence item from Labs 1-9 to every major component.

Evidence checkpoint: mark item 6 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

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104. Add a human-review decision point for low confidence, harmful content, anomaly or missing grounding.

Evidence checkpoint: mark item 7 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

105. Complete the issue register with severity, owner, workaround and acceptance decision.

Evidence checkpoint: mark item 8 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

106. Complete the security, privacy, responsible-AI, monitoring and cost controls.

Evidence checkpoint: mark item 9 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

107. Run a peer review against evidence/capstone-checklist.md and resolve all critical gaps.

Evidence checkpoint: mark item 10 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

108. Submit the evidence pack to the trainer-approved LMS location and clean up all temporary Azure resources.

Evidence checkpoint: mark item 11 in labs/lab-10-ai900-capstone-course-review/evidence/checklist.md.

Acceptance checks

- [ ] Architecture names inputs, outputs and stores
- [ ] Canonical schema and cleaning rules included
- [ ] Each component has evidence
- [ ] Human review and issue ownership are explicit
- [ ] Cleanup is confirmed

Troubleshooting and cleanup

- If the UI differs, search for the named resource/capability and record the difference.
- If access fails, verify Entra identity, role assignment, region and quota; do not expose credentials.
- If output is weak, preserve the failing input, record the expected/actual difference and route to human review.
- Retain only redacted evidence, delete or stop billable resources, and record the cleanup timestamp.

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Assessment preparation

- Written Assessment (WA-SAQ): 11 open-ended questions, 60 minutes, open book.
- Practical Performance (PP): 4 lab-aligned tasks, 90 minutes, open book.
- Instrument type, item counts, K/A mapping and durations mirror Assessment Plan v5.0.
- Assessment papers are confidential and are not included in the public GitHub repository.

Support

Course page: https://www.tertiarycourses.com.sg/wsq-microsoft-azure-ai-fundamentals-ai-900.html

LMS/TMS: https://lms-tms.tertiaryinfotech.com/

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